TRUNCATIONS PROVIDE BOUNDED MODELS FOR COHOMOLOGICALLY BOUNDED COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4001-4008, immediately following the statement at 3976-3999.

Editorial note. The three parts share one truncation choice; part (3) uses the double identity of N.

Allowed standard prerequisites (editorial).

- Canonical truncation functors. Homology, Section homology-section-truncations, cited at proof lines 4007-4008. Definition of tau truncations and their canonical maps.

Source reference. Original reference 0118: focused-seeds/geometry/S9-der-truncation-bounded-models/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $K^\bullet$ be a complex.*

(1) *If $H^n(K^\bullet) = 0$ for all $n \ll 0$, then there exists a quasi-isomorphism $K^\bullet \to L^\bullet$ with $L^\bullet$ bounded below.*
(2) *If $H^n(K^\bullet) = 0$ for all $n \gg 0$, then there exists a quasi-isomorphism $M^\bullet \to K^\bullet$ with $M^\bullet$ bounded above.*
(3) *If $H^n(K^\bullet) = 0$ for all $|n| \gg 0$, then there exists a commutative diagram of morphisms of complexes*

$$\begin{array}{ccc} K^\bullet & \longrightarrow & L^\bullet \\ \uparrow & & \uparrow \\ M^\bullet & \longrightarrow & N^\bullet \end{array}$$

where all the arrows are quasi-isomorphisms, $L^\bullet$ bounded below, $M^\bullet$ bounded above, and $N^\bullet$ a bounded complex.

Proof. Pick $a \ll 0 \ll b$ and set $M^\bullet = \tau_{\leq b}K^\bullet$, $L^\bullet = \tau_{\geq a}K^\bullet$, and $N^\bullet = \tau_{\leq b}L^\bullet = \tau_{\geq a}M^\bullet$. See Homology, Section 0118 for the truncation functors. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.